MIL-R-22578B(Wep)
19 February 1964

Superseding
MIL-R-22578A(Wep)
21 December 1961

MILITARY SPECIFICATION

RESORCINOL

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers resorcinol for use in explosives.

1.2 Classification - The resorcinol shall be of the following grades and classes specified (See Section 6.2):

Grade A - 99.5 percent minimum

Class 1 - crystal
2 - powder

Grade B - 99.5 percent minimum, Technical Grade Flake

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-D-715	Drums, Fiber with Recessed Ends
PPP-D-723	Drums, Fiber

FSC 6810

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STANDARDS

Federal

FED-STD-791	Lubricants, Liquid Fuels, and Related Products; Methods of Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government inspector. Copies of this specification and other unclassified specifications and drawings required by the contractor in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code CD1), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM-D-1500-58T	Method of Test for ASTM Color of Petroleum Products
ASTM-D-1631-59T	Water in Phenol and Related Materials by the Iodine Reagent Method

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

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Consolidated Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

American Trucking Association

National Motor Freight Classification Rules and Container Regulations

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N. W., Washington 6, D. C.)

3. REQUIREMENTS

3.1 Material - The materials used in the manufacture of the resorcinol shall be of high quality. The resorcinol shall conform to the applicable requirements as specified herein and shall be visibly free of contaminants and foreign material such as grit and dirt.

3.2 Chemical and physical properties - The chemical and physical properties of resorcinol shall conform to the requirements of Table I.

TABLE I

CHEMICAL AND PHYSICAL PROPERTIES

Property	Requirement	Test Paragraph
Resorcinol Content, % min.	99.5%	4.4.2
Ash, % max.	0.005	4.4.3
Moisture, % max.	0.2	4.4.4
Phenol	No perceptible odor	4.4.5
Catechol	No turbidity	4.4.6
pH, min.	4.4	4.4.7
Water Insoluble, % max.	Zero	4.4.8
Molten Color, Grade B, max.	1.0	4.4.9

3.3 Workmanship - The resorcinol shall be uniform in quality and shall conform to the requirements of this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Acceptance inspection - Conformance of the resorcinol to the requirements of this specification shall be determined entirely by means of acceptance inspection. The acceptance inspection shall consist of an examination for acceptability of facilities and quality control methods used by the manufacturer, examining and testing the acceptance samples (4.3.2) for all of the requirements of this specification, and an examination of the sample of filled containers (4.3.3) for conformance to Section 5.

4.3 Sampling -

4.3.1 Lot - A lot shall consist of all resorcinol of the same grade and class, manufactured by the same process, packed as specified in the contract or order (see 6.2) and offered for acceptance at one time.

4.3.2 Sample for test - From each lot offered for acceptance under contract, a random sample shall be selected in accordance with MIL-STD-105, Inspection Level II. A primary sample of one pint of material shall be removed from each container selected and stored in a large mouth glass stoppered bottle. Each primary sample shall be labeled so that the container from which it was taken can be easily identified. A composite sample shall be made of equal portions of these samples and a minimum of one pint shall be used for acceptance tests as specified in 4.4. The sample shall be labeled to show the name and address of the manufacturer, plant, name of product, specification number, lot number, date of sampling, and contract number.

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4.3.3 Sample for examination of filled containers - A random sample of filled containers shall be selected from each lot of resorcinol offered for acceptance under contract, in accordance with Standard MIL-STD-105 at inspection level I and acceptable quality level (AQL) = 2.5 percent defective, to determine compliance with the requirements of Section 5.

4.4 Inspection methods - Unless otherwise specified, all examinations and tests shall be conducted as specified herein to determine compliance with the applicable requirements of Section 3.

4.4.1 Visual inspection - Conformance of the resorcinol to the requirements for material (3.1), and workmanship (3.3) shall be determined by visual inspection.

4.4.2 Resorcinol content -

4.4.2.1 Reagents -

1. Standard sodium thiosulfate solution, 0.1N
2. Potassium bromide - bromate solution, 0.2N
3. Hydrochloric acid, C.P. (1:1 aqueous)
4. Potassium iodide solution, 15 percent
5. Starch indicator solution, 0.5 percent

4.4.2.2 Procedure - Take approximately 1.5 grams of resorcinol and weigh accurately to the fourth decimal place, dissolve in distilled water, and dilute to 500 milliliters(ml) in a volumetric flask. Mix thoroughly. Transfer by pipette 20 ml. of the sample solution to an iodine flask. Transfer 25 ml. of the bromide-bromate (0.2N) solution to the flask, dilute with 100 ml. of distilled water, add 10 ml. of hydrochloric acid solution and immediately stopper the flask. Shake the flask and allow it to stand for three minutes. Place 15 ml. of potassium iodide solution in the lip of the flask and remove the stopper just sufficiently to allow the potassium iodide solution to enter the flask without allowing any bromine vapor to escape. With the stopper in place, shake the flask thoroughly. Remove the stopper, and wash its bottom surface down the neck of the flask with 20 ml. of distilled water. Titrate the liberated iodine with the standard

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0.1N sodium thiosulfate solution, using starch indicator solution after the reaction mixture has become straw colored. Carry out a blank titration using 75 ml. of distilled water, 25 ml. of bromide-bromate solution and 10 ml. of hydrochloric acid. After the three minute waiting period, add 15 ml. of potassium iodide solution and titrate the liberated iodine by the same procedure used for the sample.

4.4.2.3 Calculation -

$$\text{Resorcinol content} = \frac{\text{ml.} \times N \times 1.835}{W}$$

Where:

ml. = ml. thiosulfate for blank - ml. thiosulfate for sample

N = normality of thiosulfate

W = wt. of sample in aliquot (1/25 of that weighed)

4.4.3 Ash - Heat a platinum, porcelain, or silica crucible to redness, cool, and weigh; reheat to constant weight so that repeated weighings are within 0.5 milligram (mg). Take a 5 to 10 gram sample and weigh to constant weight within 0.1 mg. Place approximately 1 to 5 grams of the sample in the crucible and heat gently with a Bunsen burner until the material can be ignited at the surface. Remove the burner and allow the material to burn completely. After all of the sample has burned completely, heat the residue with a strong flame or in a muffle furnace at 700° to 750°C. until all the carbonaceous material has disappeared. Cool and weigh the crucible. Reheat, cool and weigh to constant weight within 0.5 mg.

4.4.4 Moisture - The moisture shall be determined in accordance to ASTM-D-1631-59T or Fed. Test Method Std. 791, Method 3253.

4.4.5 Phenol - Dissolve one gram of the sample in 20 ml. of distilled water and warm gently. Phenol is not present if its odor is not detectable.

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4.4.6 Catechol - Dissolve one gram of the sample in 20 ml. of distilled water. Add two drops of dilute acetic acid (1:1) to 10 ml. of the solution. Mix thoroughly. Add 0.5 ml. of 10 percent lead acetate solution. No turbidity is produced if catechol is not present.

4.4.7 pH - Weigh 5 grams of the sample into a 150 ml beaker. Add 50 ml. of distilled water having a pH of 6.8 ± 0.2. Stir the mixture until the sample is dissolved and determine the pH by means of a pH meter equipped with glass-calomel electrodes.

4.4.8 Water insolubles - Weigh a 10 gram sample and dissolve in 100 ml. of distilled water, filter with vacuum through a 30 ml. medium porosity fritted disk filter and dry for 1 hour at 105° to 110°C. Calculate the percent of material insoluble in water as follows:

$$\text{Percent Water Insoluble} = \frac{\text{Weight of Residue} \times 100}{\text{Weight of Sample}}$$

4.4.9 Color - The color on Grade B shall be determined in accordance to ASTM-D-1500-58T.

4.5 Rejection criteria - If a sample fails to meet any of the test requirements of this specification, the lot represented by the sample shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - Not applicable.

5.2 Packing -

5.2.1 Level A -

5.2.1.1 Exterior containers - Packing of Grade A resorcinol shall be in 51 gallon fiber drums having a gasketed lid and plastic lining in accordance with PPP-D-715, Type II and PPP-D-723, Type III, Grade A. Each drum shall not exceed 100 pounds net. Packing of Grade B resorcinol shall be in 38 gallon fiber drums having a gasketed lid and plastic lining in accordance with PPP-D-715, Type II and PPP-D-723, Type III, Grade A. Each drum shall not exceed 200 pounds net (see 6.2).

5.2.2 Level B -

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5.2.2.1 Exterior containers - Packing of Grade A resorcinol shall be in 51 gallon fiber drums having a gasketed lid and plastic lining in accordance with PPP-D-715, Type I and PPP-D-723, Type II, Grade A. Each drum shall not exceed 100 pounds net. Packing of Grade B resorcinol shall be in 38 gallon fiber drums having a gasketed lid and plastic lining in accordance with PPP-D-715, Type I and PPP-D-723, Type II, Grade A. Each drum shall not exceed 200 pounds net (see 6.2).

5.2.3 Level C - Resorcinol shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transporation.

5.3 Marking -

5.3.1 Special markings - None, unless otherwise specified.

5.3.2 Normal markings - In addition to the markings required by contract or order, shipping containers shall be marked in accordance with the requirements of MIL-STD-129. The nomenclature shall be according to Class and Grade material, vis:

Resorcinol - Grade A, Class 1 - Crystal, 99.5 percent min.
Resorcinol - Grade A, Class 2 - Powder, 99.5 percent min.
Resorcinol - Technical "Mid Cut" Grade B - Flake, 99.5 percent min.

6. NOTES

6.1 Intended use - The material conforming to this specification is intended for use in various ordance applications.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Type and grade required

(c) Quantity required

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(d) Size and type of shipping container

(e) Level of packaging and packing desired

6.2.1 Criteria for use of proper level of preservation, packaging and packing shall be as follows:

For Level A - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B - This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or less or domestic locations for an indefinite period.

For Level C - This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 49-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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DEPARTMENT OF THE NAVY
BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

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